Portfolio Optimization Primer

Healthcare Sector | 2/05/2026

Overview of Multifactor Model and Hierarchical Risk-Parity

Risk-Parity Model

Overview of Optimization Techniques

The Risk Parity portfolio optimization uniquely focuses on equalizing the portfolio's volatility distribution rather than capital allocation. This ensures that each asset contributes equally to total portfolio volatility, providing broader exposure across economic conditions, increasing allocations to lower-volatility assets, and delivering more stable returns with reduced drawdowns. However, these updated weights may lag in performance during strong upswings in high-volatility assets like equities.

Economic Interpretation

Because of its structural design, Risk Parity exhibits sensitivities to macroeconomic conditions through its reliance on historical volatility and correlation estimates. In this healthcare portfolio, the Risk Parity optimizer increases exposure to ABBV (+2.12%), JAZZ (+1.88%), and UNH (+2.47%), while reducing allocations to MRK (-2.49%), AZN (−1.64%), and THC (−2.49%). These shifts are driven by historical volatility and covariance estimates rather than forward-looking macroeconomic or fundamental considerations. Allocation to VEEV remains relatively stable (+0.43%), reflecting minimal changes in its estimated risk contribution. Risk Parity works best in calm market environments with stable relationships between stocks, but it does not adjust for broader economic trends, regulatory changes, or individual company growth.

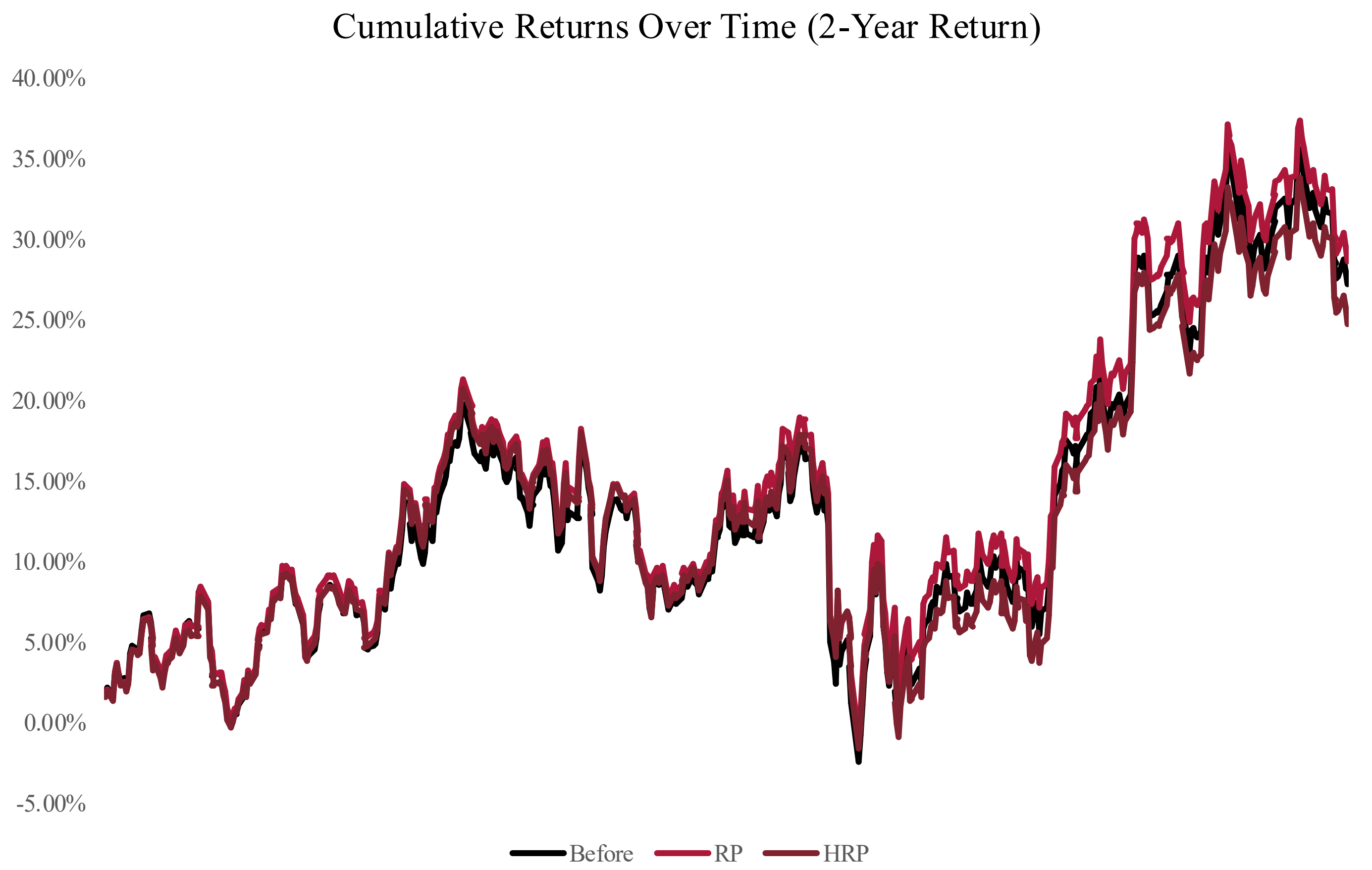
Hierarchical Risk-Parity (HRP)

Overview of Optimization Techniques

In the healthcare sector, where many equities exhibit correlated return behavior, Hierarchical Risk Parity (HRP) helps limit concentration by allocating capital across groups of correlated stocks rather than across individual securities alone. By clustering healthcare stocks based on their historical correlation structure, HRP enhances diversification relative to traditional Risk Parity. Instead of relying on precise estimates for how every pair of stocks moves together, this framework allocates risk across broader groups of similar stocks, making the portfolio more stable when data is noisy. However, HRP still depends on how stocks are grouped together, and different grouping methods can lead to different portfolio allocations.

Economic Interpretation

By clustering stocks based on their co-movement, HRP limits risk concentration across highly correlated healthcare subsectors, leading to more balanced sector exposure. In this portfolio, HRP increases allocations to ABBV (+1.30%), JAZZ (+1.69%), and UNH (+5.78%), while reducing exposure to MRK (−2.53%), AZN (−2.34%), and THC (−3.60%). These shifts reflect correlation structure rather than fundamental views, lowering overexposure to closely related pharmaceutical and provider names. Allocation to VEEV remains relatively stable (−0.31%), preserving exposure to differentiated, less-correlated growth drivers. Because HRP adjusts allocations dynamically based on changing correlation patterns instead of fixed volatility assumptions, it promotes a more resilient mix of defensive and growth-oriented healthcare stocks across varying market conditions.



Overview of Conditional Value-at-Risk and Black-Litterman

Conditional Value-at-Risk

Overview of Optimization Techniques

Conditional Value-at-Risk optimization chooses portfolio weights by prioritizing the prevention of extreme losses rather than just balancing average volatility. Instead of treating all fluctuations as risk, it isolates the worst-case scenarios for each stock and favors combinations that minimize the expected damage during market crashes. Its strengths are that it explicitly protects against tail risk events like clinical trial failures, handles skewed return patterns better than traditional models, and offers superior defense for volatile sectors. Its weaknesses are that it relies heavily on having sufficient historical crash data, ignores upside volatility in its risk calculation, and can be computationally intensive to optimize.

Economic Interpretation

Because of its focus on tail risk, Conditional Value-at-Risk (CVaR) exhibits sensitivities to extreme downside scenarios rather than average volatility. In this healthcare portfolio, the CVaR optimizer aggressively increases exposure to ABBV (+4.87%) and AZN (+3.10%), while significantly cutting THC (−5.37%) and VEEV (−3.34%). These shifts are driven by a mathematical preference for stocks that demonstrate resilience during market crashes or "fat tail" events, rather than just low daily variance. The model also increases JAZZ (+2.44%) and UNH (+1.73%), identifying them as valuable diversifiers that do not crash in unison with the broader group. CVaR works best in turbulent market environments where protecting against regulatory shocks or clinical failures is paramount, as it specifically penalizes assets that contribute to worst-case portfolio losses regardless of their upside potential.

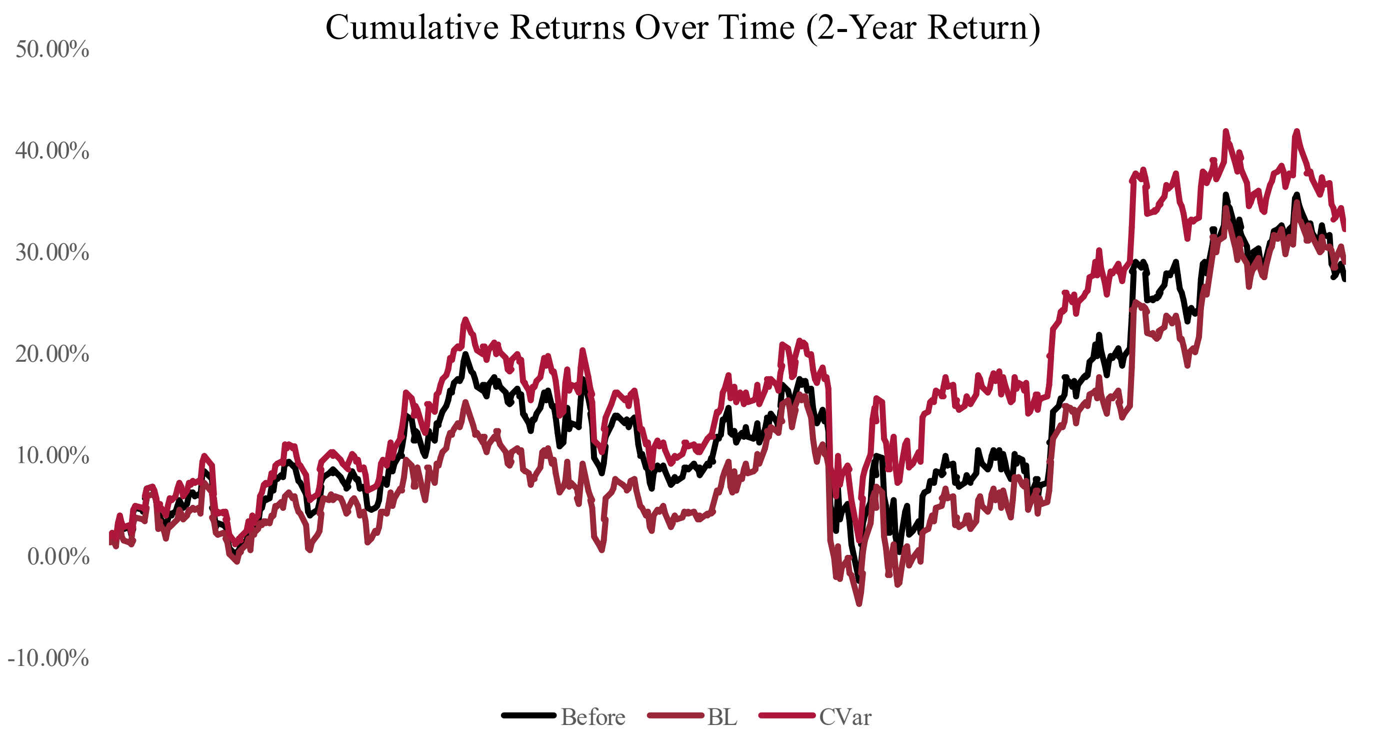
Black-Litterman

Overview of Optimization Techniques

Sentiment-Enhanced Black–Litterman builds on the standard Black–Litterman approach by combining two sources of "what we think will happen next": the team's own views and the tone of recent financial news. In practice, it reads current headlines about each company, summarizes whether the coverage is mostly positive or negative, and then uses that information to gently tilt the portfolio toward stocks with stronger sentiment and away from those with weaker sentiment, while still keeping the overall portfolio diversified. The benefit is that it makes the process less dependent on one person's opinion and more responsive to new information that may be emerging in the market. The tradeoff is that it only works well if the news is reliable and consistently available, and it adds extra moving parts that can amplify noise when headlines are misleading, overly hyped, or intentionally manipulated.

Economic Interpretation

Because of its ability to incorporate active views, Black-Litterman exhibits sensitivities to forward-looking convictions about company-specific catalysts and market trends. In this healthcare portfolio, the Black-Litterman optimizer heavily increases allocations to high-conviction names like JAZZ (+8.92%), ABBV (+4.87%), and MRK (+1.54%), pushing them effectively to their position limits. Conversely, it dramatically reduces exposure to UNH (−9.05%) and VEEV (−6.63%), reflecting a strategic pivot away from insurers and tech-valuation risks. These shifts are driven by the specific confidence levels assigned to future returns rather than historical price data alone. Black-Litterman works best in regime-shifting environments where historical data is misleading, allowing the portfolio to proactively position for specific outcomes like drug approvals or policy changes rather than reactively following past trends.

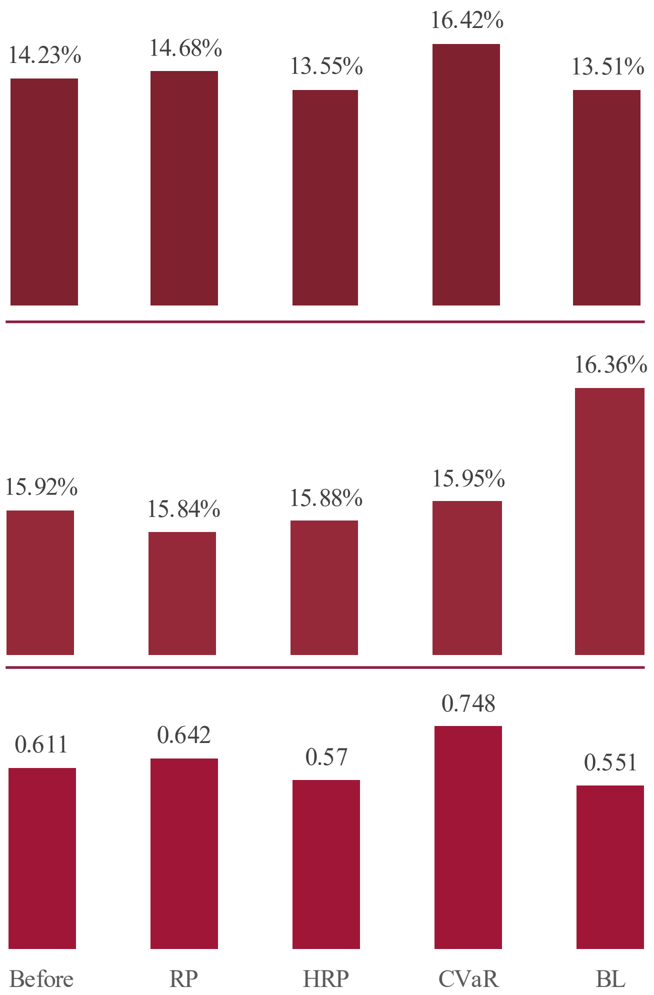


Portfolio Optimization Primer

Healthcare Sector | 2/05/2026

Comparative Analysis

Optimizer	Annual Return	Annual Vol	Sharpe Ratio	Sortino Ratio
Before	**14.23%**	**15.92%**	**0.611**	**0.880**
RP	**14.68%**	**15.84%**	**0.642**	**0.924**
HRP	**13.55%**	**15.88%**	**0.570**	**0.813**
CVaR	**16.42%**	**15.95%**	**0.748**	**1.087**
BL	**13.51%**	**16.36%**	**0.551**	**0.785**



Expected return (Annual Return)

Highest (CVaR):
Lowest (BL):

Interpretation: The CVaR optimizer achieves the highest expected return, largely driven by its more concentrated allocations that favor downside-risk control while maintaining volatility comparable to other strategies. In contrast, the Black-Litterman optimizer delivers the lowest return, reflecting its more conservative and uneven allocation structure that prioritizes stability over return maximization.

Volatility (Annual Vol)

Lowest (RP): 15.84%
Highest (BL): 16.36%

Interpretation: Volatility is tightly clustered across all strategies, indicating similar overall risk exposure. Risk Parity achieves the lowest volatility through balanced risk contributions, while Black-Litterman exhibits the highest volatility due to its broader diversification and reliance on equilibrium assumptions.

Sharpe ratio (Risk Adj)

Highest (CVaR): 0.748
Lowest (BL): 0.551

Interpretation: The CVaR optimizer delivers the strongest Sharpe ratio performance, indicating superior compensation for total risk taken. In contrast, the Black-Litterman approach produces the weakest Sharpe ratio return, reflecting lower efficiency in converting risk into excess return over the evaluation period.

Analysis:

After analyzing the optimization models, ABBV and MRK had the highest average weights at 18.42% and 16.73%, with ABBV receiving consistent 20% allocations across CVaR and Black-Litterman models. VEEV had the lowest average at 11.35%, being heavily limited in the Black-Litterman model (7.18%) likely due to weaker market equilibrium or sentiment-based expectations. UNH showed significant variability, with a high HRP allocation (17.56%) but near exclusion from Black-Litterman (2.73%), indicating strong return potential but higher volatility or correlation risks that certain frameworks penalize.

Capital Allocation For Each Stock

Tickers	Before	RP	HRP	CVaR	BL
ABBV	15.13%	17.25%	16.43%	20%	20%
JAZZ	11.08%	12.96%	12.77%	13.52%	20%
THC	14.51%	11.75%	10.91%	9.14%	17.10%
MRK	18.46%	15.97%	15.93%	15.02%	20.0%
UNH	11.78%	14.25%	17.56%	13.51%	2.73%
VEEV	13.81%	14.24%	13.50%	10.47%	7.18%
AZN	15.23%	13.59%	12.89%	18.33%	12.99%

Recommended Optimization Framework: Conditional Value-at-Risk

A robust optimization framework utilizing Conditional Value-at-Risk (CVaR) is applied to the healthcare sector portfolio, providing both defensive stability and protection against extreme market events. This approach uses historical data to identify the worst-case loss scenarios, preventing the optimizer from underestimating the danger of sudden stock crashes common in biopharma. Complementing this, the optimization algorithm minimizes the expected loss during these tail events while satisfying return targets, ensuring capital preservation is prioritized over aggressive volatility harvesting. This method combines the statistical rigor of tail risk analysis with the practical need for drawdown control, enabling the portfolio to navigate regulatory cliffs and clinical trial failures while avoiding the ruinous losses that standard mean-variance models often fail to predict.

Fit with the Nature of Healthcare Businesses

This framework fits healthcare businesses particularly well because the sector is defined by binary risks, including patent expirations (ABBV), clinical trial results (JAZZ, AZN), and policy shifts (UNH, THC). A standard model might incorrectly treat these risks as standard daily noise, but the CVaR model isolates their potential for catastrophic downsides. This is reflected in our allocation table, where the CVaR model maintains high conviction (20.00%) in stable pharmaceutical giants like ABBV while intelligently sizing binary-risk names like JAZZ (13.52%) and THC (9.14%) based on their contribution to portfolio tail risk. This mirrors economic reality, as drug pipeline failures and reimbursement cuts create skewed return distributions. The CVaR component leverages this clarity to maximize exposure to diversifiers without unknowingly concentrating risk in a single regulatory outcome.

Fit with the Macro Environment

The healthcare landscape is currently marked by strong regulatory oversight and company-specific pipeline volatility. Standard investment formulas struggle here because they assume returns follow a normal bell curve, often underestimating the frequency of sharp sell-offs caused by FDA rejections or legislative changes. Our CVaR approach fixes this by focusing specifically on the left tail of the return distribution, ignoring the noise of average trading days to solve for survival during crises. This helps us identify resilient portfolio structures rather than just chasing low-volatility illusions. The result is a portfolio that earned over 16% by defending against sector-specific shocks while participating in the growth of successful drug launches.

Practical Implications for Portfolio Construction

In practical terms, this framework improves position sizing by determining weights through a "loss contribution" lens before optimizing for returns. For example, where a Risk Parity model might overweight a volatile name to balance variance, the CVaR model retains a measured 13.52% position in JAZZ and 10.47% in VEEV. It recognizes that their potential downsides are driven by specific factors distinct from UnitedHealth or Merck, offering true downside diversification. From a risk management standpoint, the framework prevents the optimizer from ignoring the magnitude of potential losses, resulting in a portfolio that captures the steady compounding of winners like AZN (18.33%) while strictly limiting exposure to correlated crash risks. This tail-risk approach grounds allocations in stress-test performance rather than average daily volatility alone. This results in a Sharpe Ratio of 0.748 that balances defensive stability with upside potential.

Limitations and Caveats

However, there are important limitations to consider. The CVaR model depends heavily on the availability of historical crash data, meaning if a specific type of regulatory shock has not occurred in the past, the risk estimation will be fundamentally flawed. The model is also sensitive to the chosen confidence interval, so selecting an overly narrow threshold might miss moderate but frequent drawdowns that erode value over time. Finally, this approach depends on more complex optimization methods and assumes that extreme market relationships will continue, an assumption that can break down during major unexpected events such as a global pandemic or significant policy changes.

CVaR Weightings	
ABBV	20%
JAZZ	13.52%
THC	9.14%
MRK	15.02%
UNH	13.51%
VEEV	10.47%
AZN	18.33%

Performance	Annual Return	Annual Vol	Sharpe Ratio	Sortino Ratio
CVaR	16.42%	15.95%	0.748	1.087